

RIO CUARTO, Argentina

WMO: 874530

Lat: 33.0956S

Lon: 64.2769W

Elev: 1381

StdP: 13.98

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.4	35.2	13.7	11.6	50.8	17.5	14.0	49.7	32.6	59.1	28.9	58.6	5.8	270	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	93.2	69.3	90.4	68.8	87.9	68.1	74.9	85.3	73.3	83.2	71.8	81.8	16.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.9	124.4	79.8	70.2	117.0	78.4	68.5	110.4	76.9	39.6	85.5	38.0	83.5	36.6	82.1	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	26.7	23.3	26.7	99.5	4.2	3.6	23.7	102.1	21.2	104.2	18.9	106.2	15.8	108.8		
(5)			24.2	78.9	3.6	2.0	21.6	80.3	19.5	81.5	17.5	82.7	14.9	84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	74.1	71.7	68.7	62.6	56.5	51.0	49.9	53.5	58.3	63.8	69.0	72.9
	DBStd	10.54	5.19	5.90	5.98	6.77	6.57	5.49	6.36	7.53	7.58	7.07	6.42	5.85
	HDD50	214	0	0	0	3	15	53	81	45	16	1	0	0
	HDD65	2072	3	12	32	122	275	421	470	366	224	107	32	8
	CDD50	4816	746	608	579	381	217	82	77	154	264	430	570	708
	CDD65	1202	284	199	146	51	12	0	1	11	23	71	153	251
	CDH74	10882	2529	1606	1100	412	76	5	17	158	293	761	1517	2408
	CDH80	3943	992	568	327	84	7	0	3	49	79	260	571	1003
Wind	WSAvg	10.7	10.8	10.0	9.7	9.5	8.8	8.8	10.0	11.3	12.6	13.0	12.4	11.8
Precipitation	PrecAvg	31.8	5.1	3.9	4.3	2.2	1.1	0.6	0.5	0.6	1.6	3.3	4.8	5.4
	PrecMax	44.5	11.6	14.0	18.3	5.6	8.6	4.4	4.4	3.3	6.1	10.5	9.1	12.8
	PrecMin	18.4	0.4	0.9	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.1	1.4	1.1
	PrecStd	6.0	2.6	2.6	2.9	1.4	1.5	0.8	0.7	0.7	1.3	2.0	2.2	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.5	94.3	91.1	87.2	80.7	73.7	75.4	86.1	87.7	92.9	94.1	96.9
		MCWB	70.6	71.4	70.0	67.9	64.8	57.6	57.8	62.7	62.1	67.2	66.6	70.9
	2%	DB	91.7	89.8	87.3	82.2	75.6	68.5	69.7	77.9	81.3	86.4	89.9	92.8
		MCWB	71.0	71.5	70.2	65.8	62.5	54.7	54.8	58.5	59.9	63.5	65.5	69.2
	5%	DB	88.7	86.4	83.9	78.9	71.5	64.9	65.7	71.8	76.9	82.0	86.3	89.5
		MCWB	70.3	70.8	68.8	63.7	59.7	52.6	52.2	54.2	58.0	61.9	64.5	68.0
	10%	DB	85.6	83.1	80.6	75.3	67.8	61.9	61.4	67.6	72.6	77.4	82.7	85.9
		MCWB	69.4	69.7	67.5	62.8	58.1	51.5	50.1	52.7	55.5	59.3	63.2	67.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	77.2	75.6	72.8	67.5	62.6	62.5	65.7	66.5	70.8	72.3	75.4
		MCDB	86.0	87.4	85.6	80.7	75.6	69.4	71.0	79.6	80.4	81.9	84.0	89.0
	2%	WB	74.6	74.7	72.8	69.4	64.8	58.6	58.0	61.2	62.8	67.1	69.5	72.9
		MCDB	84.7	83.7	81.5	75.9	72.5	63.5	64.1	72.9	74.1	79.8	81.5	85.9
	5%	WB	72.8	72.9	70.9	66.9	62.5	56.1	55.3	57.6	60.3	64.7	67.4	70.9
		MCDB	83.6	81.2	79.6	74.8	68.6	60.9	60.4	67.4	71.9	76.0	80.0	83.2
	10%	WB	71.2	71.3	68.9	64.8	60.1	53.6	52.5	54.4	57.9	62.4	65.6	69.2
		MCDB	82.0	79.9	77.4	72.5	65.4	58.9	59.0	65.0	68.8	73.4	77.8	81.0
Mean Daily Temperature Range	5% DB	MDBR	19.3	18.2	18.6	18.6	17.8	19.3	19.8	21.6	21.7	20.9	21.3	20.8
		MCDBR	23.8	22.4	22.9	24.9	23.3	25.4	26.8	29.3	28.6	27.5	27.0	26.0
		MCWBR	9.9	9.1	10.1	11.5	12.1	14.1	14.5	14.3	13.4	12.0	11.0	10.3
	5% WB	MCDBR	20.4	18.8	19.6	19.6	18.6	19.4	20.4	25.0	23.5	22.9	22.7	21.8
		MCWBR	10.7	10.2	9.9	10.9	11.0	12.9	13.3	13.9	12.6	12.4	11.2	10.8
Clear-Sky Solar Irradiance	taub	0.386	0.378	0.362	0.362	0.330	0.310	0.322	0.356	0.404	0.389	0.380	0.385	
	taud	2.374	2.407	2.453	2.418	2.459	2.515	2.441	2.325	2.189	2.293	2.352	2.366	
	Ebn at Noon	302	298	291	272	266	265	265	269	272	291	302	304	
	Edh at Noon	41	38	34	32	27	24	27	34	44	43	42	42	
All-Sky Solar Radiation	RadAvg	2298	1979	1663	1214	897	835	897	1196	1554	1845	2187	2341	
	RadStd	140	156	145	151	99	92	73	101	135	172	140	157	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.12	+0.90	+1.31	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	+1.01	+1.12	+1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page